SpThreshold: Replication Threshold for Bayesian Multilevel Models of Areal Data

Statistical Model

$$Y_{ij} = \beta_0 + \sum_{k=1}^{p} \beta_k \mathrm{x}_{kij} + \theta_i + \varepsilon_{ij}, \quad i = 1, \ldots, n;\ j = 1, \ldots, m_i.$$

- n: Number of spatial areas (e.g., counties, census tracts);
- m_i: Number of observations within area i;
- p: Number of covariates;
- x_{kij}: Value of covariate k for observation j in area i;
- θ_i: Random intercept shared by observations in area i;
- ε_{ij}: Observation-level residual error.

$$\varepsilon_{ij} \mid \sigma^2 \overset{\text{iid}}{\sim} \mathrm{N}(0, \sigma^2).$$

For the spatial model,

$$\theta_i \mid \boldsymbol{\theta}_{-i}, \tau^2_{\text{sp}}, \rho \sim \mathrm{N}\left(\frac{\rho \sum_{i'=1}^{n} \mathrm{w}_{ii'} \theta_{i'}}{\rho \sum_{i'=1}^{n} \mathrm{w}_{ii'} + 1 - \rho}, \frac{\tau^2_{\text{sp}}}{\rho \sum_{i'=1}^{n} \mathrm{w}_{ii'} + 1 - \rho} \right).$$

- $\boldsymbol{\theta}^{\mathrm{T}}_{-i} = (\theta_1, \ldots, \theta_{i-1}, \theta_{i+1}, \ldots, \theta_n)$;
- $\mathrm{w}_{ii'}$: Equal to one if areas i and i' are neighbors, zero otherwise; $\mathrm{w}_{ii} = 0$;
- ρ: Spatial correlation parameter, with $0 \leq \rho < 1$;
- Equivalent joint form: $\boldsymbol{\theta} \mid \tau^2_{\text{sp}}, \rho \sim \mathrm{N}_n(\mathbf{0}_n, \tau^2_{\text{sp}} Q(\rho)^{-1})$, where $Q(\rho) = \rho \mathrm{L} + (1 - \rho) I_n$ and $\mathrm{L} = \mathrm{diag}(\mathrm{w}_{1\cdot}, \ldots, \mathrm{w}_{n\cdot}) - \mathrm{W}$ is the graph Laplacian.

For the nonspatial model,

$$\theta_i \mid \tau^2_{\text{ns}} \overset{\text{iid}}{\sim} \mathrm{N}(0, \tau^2_{\text{ns}}).$$

The random intercepts and residual errors are independent of the covariates and of each other.

Prior Information

$f(\beta_k) \propto 1, \quad k = 0, \ldots, p;$

$\sigma^2 \sim \text{Inverse Gamma}(a_{\sigma^2}, b_{\sigma^2});$

- Default setting: $a_{\sigma^2} = 0.01$, $b_{\sigma^2} = 0.01$.

The random-intercept variance has the following prior under the corresponding model:

$\tau^2_{\text{ns}} \sim \text{Inverse Gamma}(a_{\tau^2}, b_{\tau^2});$

$\tau^2_{\text{sp}} \sim \text{Inverse Gamma}(a_{\tau^2}, b_{\tau^2});$

- Default setting: $a_{\tau^2} = 0.01$, $b_{\tau^2} = 0.01$.

Inverse-gamma priors use shape a and scale b, with density proportional to $t^{-a-1}\exp(-b/t)$.

For the spatial model,

$\rho \sim \text{Beta}(a_\rho, b_\rho)$;

- Default setting: $a_\rho = b_\rho = 1$, giving a uniform prior on $(0, 1)$.

Default Initial Values

- $\beta_k = 0$ for $k = 0, \ldots, p$;
- $\theta_i = 0$ for all i;
- $\sigma^2 = 1.00$;
- $\tau^2_{\text{ns}} = 1.00$ or $\tau^2_{\text{sp}} = 1.00$, according to the selected model;
- $\rho = 0.50$ for the spatial model.

Model Indicator

- `model_indicator = 0`: Nonspatial model with independent Gaussian random intercepts;
- `model_indicator = 1`: Spatial model with a Leroux conditional autoregressive prior.

MCMC Sampling

For Gaussian outcomes, posterior sampling proceeds in the following order within each iteration:

1. $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_p)^{\text{T}}$ (Gibbs);
2. $\boldsymbol{\theta}$ (Gibbs);
3. τ^2_{ns} or τ^2_{sp} (Gibbs);
4. σ^2 (Gibbs);
5. ρ for the spatial model (Metropolis step with a random-walk proposal on the logit scale).

The proposal standard deviation for ρ is adapted during burn-in, targeting an acceptance rate of 0.234, and held fixed thereafter.

Random intercepts are represented with zero sum, with their mean absorbed into the intercept. Under the flat intercept prior, this preserves the slope and covariance parameter posteriors of the unrestricted model.

Derivations of the full conditional distributions are provided in the Supplementary Material. The function `posterior_summary()` provides posterior means, medians, variances, and credible intervals.

Variance Calibration

For variance comparisons, the default calibration matches the average marginal random-intercept variance between the two models:

$$\tau^2_{\text{ns}} = \frac{\tau^2_{\text{sp}}}{n}\,\text{tr}\{Q(\rho)^{-1}\} = \frac{\tau^2_{\text{sp}}}{n}\sum_{i=1}^{n}\frac{1}{1-\rho+\rho\lambda_i},$$

where λ_i are the eigenvalues of L.

- Supply `tau_ns_2` to calculate the corresponding spatial scale `tau_sp_2`;
- Supply `tau_sp_2` to calculate the corresponding nonspatial variance `tau_ns_2`;
- If both are supplied, their values are used as given.

This calculation is provided by `variance_calibration()`.

Replication Threshold m^*

For Gaussian outcomes, `m_star()` computes approximate and exact replication thresholds based on full conditional coefficient variances. These variances integrate out the random intercepts and condition on the covariance parameters and remaining regression coefficients.

Under balanced replication with m observations per area, the approximate threshold for a centered, population-standardized covariate is

$$m^*_{\text{approx}} = \max\left\{2,\left\lceil\frac{\sigma^2\left|\sum_{i=1}^{n} d_i\left\{\frac{1}{\tau^2_{\text{ns}}}-\frac{1-\rho+\rho\lambda_i}{\tau^2_{\text{sp}}}\right\}\right|}{\gamma\left(n-\sum_{i=1}^{n} d_i\right)}\right\rceil\right\}.$$

- λ_i: Eigenvalues of the graph Laplacian L, with corresponding orthonormal eigenvectors $\boldsymbol{u}_i$;
- $d_i = (\boldsymbol{u}_i^{\text{T}}\bar{\boldsymbol{x}})^2$: Squared projection of the covariate's area means onto the ith eigenvector;
- $\bar{\boldsymbol{x}} = (\bar{\text{x}}_1,\ldots,\bar{\text{x}}_n)^{\text{T}}$, where $\bar{\text{x}}_i = m^{-1}\sum_{j=1}^{m}\text{x}_{ij}$;
- The covariate is centered and standardized to population variance one, with positive within-area variation so that $n-\sum_i d_i > 0$;
- γ: Tolerance on the absolute relative variance difference, using the nonspatial variance as the denominator. Default setting: $\gamma = 0.05$.

The approximation follows from the leading $O(m^{-1})$ term of the relative variance difference and is calculated with `method = "approx"`.

The exact threshold is the smallest integer $m \geq 2$ for which the tolerance holds at every larger integer sample size. It is calculated with `method = "exact"`, using the exact variance expressions and a finite-sample bound to limit the numerical search. The calculations hold the covariate's area means and within-area variation per replicate fixed as m varies.

With multiple covariates, the comparison applies separately to each coefficient conditional on the others. The function `variance_comparison()` evaluates the full conditional variances at an observed design, including unequal replication.

Two-Stage Procedure

The function `two_stage_estimate()` applies the criterion when covariance parameters are unknown:

1. Fit a nonspatial random-intercept model by restricted maximum likelihood to estimate σ^2 and τ^2_{ns};
2. Estimate ρ using a Gaussian working likelihood for the predicted random intercepts, accounting for shrinkage, observation noise, and fixed-effect estimation, and obtain τ^2_{sp} through variance calibration.

The function returns the covariance parameter estimates and full conditional variance comparisons for each predictor at the observed design.

Non-Gaussian Outcomes

The package also fits the following models using Pólya–Gamma augmentation:

- Bernoulli: Logit link;
- Binomial: Logit link, with the number of trials supplied through `trials`;
- Negative binomial: Log link, with fixed positive integer size r supplied through `nb_size`. For conditional mean μ_{ij}, the conditional variance is $\mu_{ij} + \mu_{ij}^2/r$.

Regression coefficients have independent $\text{N}(0, 10^2)$ priors by default. Prior means and standard deviations are specified through `beta_prior_mean` and `beta_prior_sd`.

The argument `center` determines whether the random intercepts are constrained to sum to zero; its default is `TRUE`.

Reference

Lin, S. and Warren, J. L. (2026+). On the Need for Spatial Random Effects in Bayesian Regression Models for Multilevel Areal Data.